

SQL & DATABASES · DAY 1 OF 10

What is a database?

Tables, rows, columns and your first SQL questions

Monday 5 October 2026

Today

- After today you can:
 - Explain what a database is and why we use one
 - Know the words table, row, column, data type and primary key
 - Find your way around DB Browser for SQLite
 - Ask the database questions with SELECT, WHERE, ORDER BY and LIMIT

Today's schedule

Time	What we do
09:00-09:10	Welcome
09:10-09:50	Theory (slides)
09:50-10:00	Break
10:00-10:55	Code-along, part 1
10:55-11:05	Break
11:05-12:00	Code-along, part 2
12:00-13:00	Lunch
13:00-14:30	Lab, part 1
14:30-14:45	Break
14:45-16:15	Lab, part 2
16:15-17:00	Walkthrough and wrap-up

What is a database?

- A place where a computer keeps information, neat and organised
- The information stays there, even when you turn the computer off
- Many people and programs can use it at the same time
- Every app you use has one: Spotify, BankID, Blocket, Instagram

Why not just use Excel?

	Excel	Database
Size	Gets slow with many rows	Millions of rows, no problem
Many users	One person edits at a time	Thousands at the same time
Rules	You can type anything anywhere	Rules stop wrong data
Questions	Click, filter, sort by hand	One line of SQL

A table is like a class list

- A database is made of tables
- A table holds one kind of thing: customers, products, orders
- It looks a bit like a sheet in Excel

customer_id	first_name	last_name	city
1	Anna	Lindqvist	Uppsala
2	Erik	Johansson	Stockholm
3	Sara	Ahmed	Göteborg

Rows and columns

- A row = one thing (one customer: Anna)
- A column = one piece of information about every thing (everyone's city)
- Every row has the same columns

Row = one person. Column = one question about everyone.

Data types: what kind of information?

Type	What it holds	Example
INTEGER	Whole numbers	42, 7, 0
REAL	Numbers with decimals	599.50
TEXT	Words and letters	'Uppsala'
NULL	Nothing / unknown	We don't know the city

Primary key: everyone needs a unique ID

- A primary key is a column that is different for every row
- It's like your personal number (personnummer)
- Two customers can both be called Anna Lindqvist
- But they can never have the same customer_id

What is SQL?

- SQL = Structured Query Language
- It's the language we use to talk to a database
- You say WHAT you want, not HOW to find it
- Pronounced 'S-Q-L' or 'sequel'. Both are fine!

```
SELECT first_name, city  
FROM customers  
WHERE city = 'Uppsala';
```

Our database this course: a webshop

- customers: who buys things
- products: what we sell
- Later we will add orders: who bought what, and when
- We use the same webshop every day, and it grows

Our tool: DB Browser for SQLite

- SQLite = a small database that lives in one file (webshop.db)
- DB Browser = a program to look inside that file
- Free, simple, and no server needed
- Tab 'Execute SQL' is where we write our questions

The SELECT recipe

- Always in this order!
- Only SELECT and FROM are required

```
SELECT    which columns?  
FROM      which table?  
WHERE     which rows?  
ORDER BY  in what order?  
LIMIT     how many?
```

Golden rules for today

- Text goes in 'single quotes': 'Uppsala'
- Numbers have no quotes: 500
- End each question with a semicolon ;
- Read the error message. It's trying to help you!

Remember

- A database keeps organised information in tables
- Row = one thing, column = one piece of information
- A primary key gives every row a unique ID
- `SELECT ... FROM ... WHERE ... ORDER BY ... LIMIT`
- NULL means 'unknown'. Use `IS NULL`, never `= NULL`

Your turn (1/2)

1. Show the name and price of all products. (12 rows)
2. Which customers live in Stockholm? (2 rows)
3. Which products cost more than 1000 kr? (3 rows)
4. Show all Accessories, cheapest first. (4 rows)
5. Which products have fewer than 10 in stock? (4 rows)

Your turn (2/2)

6. Which customers have a first name that starts with E? (2 rows)
7. Which customers joined during 2024? (3 rows)
8. Show the 5 cheapest products. (5 rows)
9. Which categories exist? Each should appear only once. (3 rows)
10. Challenge: Clothing that is in stock ($\text{stock} > 0$) and costs under 600 kr, most expensive first. (4 rows)

Coming up tomorrow

**Today we asked questions to tables
someone else built. Tomorrow you build
tables yourselves!**